

2026 DCSS 307

Computer Network

- Welcome!

Byungjin Cho

Agenda in 1st week

- **General administration**

Instructors + SEMO

- Responsible teacher: **Byungjin Cho**
 - ✓ Homepage: <https://byungjincho.github.io>
 - ✓ Email: ByungjinCho@korea.ac.kr
 - ❑ Use prefix “[ComputerNetwork] title”
 - ✓ Office: room 329A in S&T building II
 - ❑ Friday by appointment
- Assistant:
 - ✓ **임도연**, imyryryr@korea.ac.kr
- Mentors:
 - ✓ **김성현**, honeybeelawn@korea.ac.kr
 - ✓ **최민규**, minkyuchoi0320@gmail.com
- SEMO **Pre-class** (Anytime before Thu, 10:00)
 - ✓ Online lecture
 - Taking it beforehand
 - ✓ Small Group activity
 - Each group will run the self-meeting supervised by mentors in Zoom online between 11:00 – 11:40 Every Thursday (from 2nd week).
- SEMO **In-class** (Fri, 09:00-09:50)
 - ✓ Offline, room 310 in S&T building II (Default)
 - ✓ Also it can be held in Online Zoom meeting (when needed)

Course organization & learning methods

- Weekly material distribution (**PPT slide deck**)
- Pre-class Lectures online
- Pre-class small group activity
- In-class lecture
- **Project**
- Tests (Mid-term **exam** + Final **exam**)
- Learning through interactive lectures, discussion, assignments, and group works

Schedule

Week 1	Module 0 Intro	Week 9	Module 3 Network
Week 2	Module 1 Application	Week 10	Module 4 Link
Week 3		Week 11	
Week 4	Module 2 Transport	Week 12	Module 5 Case study project
Week 5		Week 13	
Week 6	Module 3 Network	Week 14	
Week 7		Week 15	
Week 8	Exam	Week 16	Exam

Learning outcomes

- Understand the structure, functions, and key protocols of computer networks
- Identify and explain how Internet protocols interconnect to provide end-to-end services
- Do Logical reasoning and improve practical skills for network system development and operation
- Explore emerging networking technologies and convey technical insights

Lecture Materials

- Main Textbook:
 - ✓ Kurose & Ross, Computer Networking: A Top-Down Approach:
Focuses on the **structured, protocol-layer model** of networks
- Lecture Slides:
 - ✓ Tailored from official textbook slides
- Additional References:
 - ✓ An Introduction to Computer Networks, P. Dordal
<https://intronetworks.cs.luc.edu/current/html/>
 - ✓ Data and Computer Communications – William Stallings
 - ✓ Computer Networks – Andrew S. Tanenbaum
 - ✓ Data Communications and Networking – Behrouz Forouzan

Assessment

- Exams (60%)
- Project (30%)
- Attitude (10%)
- Bonus points (will be informed later)

Exams (60%)

- Mid-test (30%): Closed-book in-class written test (in-person) during mid-term exam period in week 8
- Final-test (30%) : Closed-book In-class written test (in-person) in week 16 (or in week 15, will be informed later depending on the students' presentation schedule in Module 5)

Project (30%)

Overview

Explore emerging networking domains, **not fully covered in the textbook**, but reflect real-world **trends, industry** relevance, and **job interview-ready** knowledge.

Assessment Item		Description
Group (15%)	Midpoint Report	Quality of the selected topic, depth of problem definition, identification of credible references, and the clarity of the plan. Individual roles.
	Presentation (15%)	Clarity and professional delivery of the pre-recorded video (online), and the ability to defend technical choices during the live Q&A (offline).
Individual (15%)	Engagement (5%)	Basically attendance and activity in Module 5 (Attendance at others' presentations with question as well as own's presentation, and questions)
	Portfolio (10%)	Quality of technical documentation on your GitHub README.md. Must effectively document your presentation contents (Problem, Solutions, Trade-offs, etc.) formatted as a professional Tech Blog, including referenced graphs, proper citations, and logical deductions if possible
	Reflection Report (bonus)	Reflect on your learning on the project experience and overall course feedback (max 1 page)

Project

Topic scope - Pick a Network Environment

1. **High-Mobility Networks:** Vehicular/V2X, UAV/Drone swarms,
 2. **Low-Power / Massive Networks:** IoT, Smart devices, Backscatter,
 3. **Latency-Critical Networks:** Edge-assisted computing, AR/VR,
 4. **Extreme / Long-Delay Networks:** LEO Satellites/Starlink, Marine/Underwater,
 5. **High-Throughput / Broadband Networks:** 5G/6G Cellular applications,
- **Or... Any other environment!** Needs be approved by the instructor.

Project

Why Portfolio!

- In the industry, diagnosing a network bottleneck is a highly valued skill.
 - ✓ This project proves to recruiters that you can logically diagnose protocol-level issues, not just write simple code.
- Tech Blog Style Documentation:
 - ✓ Do NOT just upload a PPT or PDF file. Dead files show no real engineering documentation skills.
 - ✓ Write a beautiful README.md file, e.g., using Markdown.
- Proof of Technical Reasoning:
 - ✓ Your GitHub repository will showcase your ability to research IEEE papers, extract meaningful data, cite sources properly, and explain the "Why & How."
 - ✓ Clear technical writing is one of the most highly sought-after skills in a software engineer.

Project

Recommended Presentation Structure (1/2)

1. Introduction & Motivation

- ✓ Understand the Problem: What is the problem?
- ✓ Why does it matter in this specific domain?

2. Problem Statement

- ✓ Define the specific bottleneck.
- ✓ What is the "fatal weakness" (target metric) you are focusing on?

3. Solutions & Technical Approaches

- ✓ Investigate the Solution: What is the current or proposed solution?
- ✓ How does it work at the protocol level?
- ✓ What is the most popular or effective technique today?

Project

Recommended Presentation Structure (2/2)

4. Industry Applications, Trade-offs & Comparisons
 - ✓ Analyze Real-World Applications: Which companies are leading in this area?
 - ✓ What are they doing differently — and why?
 - ✓ What are the trade-offs of their approaches? (Compare with your empirical data!)
5. Conclusion & Future Directions
 - ✓ Reflect and Look Ahead: What are the remaining challenges or limitations?
 - ✓ What could be the next solution or future direction?

Project

Language Guideline

- As this is a major course conducted in English (though not an English language course), your **project presentations will also be delivered in English.**
 - ✓ Offline Q&A sessions may be conducted in either Korean or English, in consideration of international students.
- **Your English fluency is not being graded.** The focus is on **how clearly and logically you can explain technical content.**
 - ✓ **Don't feel pressured** — see this as a good chance to practice the kind of communication expected in job interviews or academic presentations.
 - ✓ Present confidently and clearly — that's what matters most.

Integrity

- Teamwork & Attitude Matter
 - ❑ Be polite and collaborative with your teammates. Your attitude and contribution are being monitored. These behaviors reflect the standards expected in your future career
- Use of Generative AI (GenAI) Tools
 - ❑ GenAI trained using large language models (LLM) or other methods to produce text, images, music, or code, like ChatGPT, Gemini, or GitHub CoPilot, **may be used in this course with proper documentation, citation, and acknowledgement.**
 - ❑ However, do not copy-paste content blindly. Your submissions must reflect your own understanding and effort — I may ask you to explain your work directly.
 - ❑ Always cite any external sources or tools used (but do not blindly believe references from ChatGPT)
 - ❑ If inappropriate use (e.g., full AI-generated content with no understanding) is observed, you may receive zero points for the assignment or project

Attendance policy

- Attendance **will not affect grades** directly
 - ☐ Except for small group activities and project relevant activities
 - ☐ Except for many absents
- University regulations apply to attendance for **granting** grades
 - ☐ Students absent for more than $\frac{1}{3}$ of total class hours will not receive a grade.
 - ☐ Even with approved absences (e.g., medical, official), a grade will be given only if the student attends at least $\frac{1}{2}$ of total class days.
 - ☐ Attendance must be checked every lecture.

Questions?

Q&A 게시판

Search result 0

No

Please fill-in the mandatory Survey!!!